



January 2016

Team: SC, RM, LC (75%), NW (80%)

Strategy

- [SC, LC] Quarterly call update
- [SC, ES] Refined Investment process slides
- [SC] Conversations: Skybridge, Freddie Mac
- [SC] IO Beta Update

Modeling

- **Baton** [LC]: Agency prepayment model
 - Improved CQ/CR fits
 - Investigate enhancing existing algorithms for identifying HARPed loans
- **House Price Model** [SC, CL]:
 - Developed framework for a house price model
- **Scale** [RM, NW]
 - Completed major refactoring of Scale
 - Instrument (moved CMO and Bond to instrument module)
 - PrepayModel (separated this module from the Rate Models)
 - RateModels (refactored Current Coupon Model, Mortgage Spread Model, Mortgage Rate Model and Hull White Model)
 - HpaModel (added constant appreciation and constant index projections)
 - ScaleUtilities (added simulator and ratePath to utilities)
 - ScaleFacade (added tests for: HPA Projection, CMO Replines, Mortgage Rate Projection)
 - Other infrastructural changes:
 - More consistently using QuantLib Date class
 - Switched to using QuantLib time series class
 - Switched to using mersenne twister random number generation (QuantLib)
 - Removed dependence on external library AlgLib (source of Spline routines)
 - Tied out spline algorithm with R spline algorithm



February 2016 Pipeline

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Strategy

- [SC] Document and enhance Market Risk Management framework for Master Fund
- [SC] IO Basis Report

Modeling

- **Baton** [RM, LC]: Infrastructure and fitting of prepayment-model
 - Mortgage rate model for GNMA's
 - MI Changes in database
 - Estimate GNMA Prepayment model
 - Extend conventional prepayment model to incorporate difference S-curves for HARPed versus non-HARPed loans.
- **HARP Identification Model** [LC]
 - Design algorithm for identifying HARPed loans
 - Implement algorithm for identifying HARPed loans in loan and pool database (collaboration with Technology)
- **Scale** [RM, NW]
 - Validate interest rate model with Chunnip
 - Incorporate new prepayment model into Scale
 - Generate prepay tracking report
 - Return nominal measures when calculating OAS
 - Include other risk measures like duration, convexity etc into Scale
 - Generate IOS benchmark report
 - Continue Scale refactoring
 - CashflowEngine
 - Calculator